ANGELA FOO

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EDUCATION

California Polytechnic State University, San Luis Obispo, CA **Expected December 2026**
B.S in Manufacturing Engineering **Major GPA: 3.7**

WORK EXPERIENCE

EBR SYSTEMS INC.
June 2023 - Present
Manufacturing Engineering Intern (Remote) | Sunnyvale, CA

- Design fixtures and tooling in SolidWorks — validate designs through SLA/FDM to cut prototype lead times by 2 weeks
- Manage documentation for 3 production test fixtures in PLM supporting accurate reporting and regulatory compliance to ISO 13485 for quality audits
- Led the transition of a medical-grade silicone molding process from an external supplier to in-house production, using Design of Experiment (DOE) trials to optimize process parameters
    - Reduced part lead time time by 3 weeks and costs by 20%
    - Authored work instructions to degas, cure, demold and inspect for defects
    - Optimized the process for ergonomics to reduce technician fatigue
- Qualified new crimp tooling for catheter subassembly using Minitab correlation analysis across 5 crimp diameters, establishing a process window that met tensile-strength requirements while preserving mandrel pass-through
- Developed and executed a test and inspection plan to diagnose subassembly leakage
    - Confirmed the root cause as a design issue and documented the findings in a Quality Review Event (QRE)
- Modeled a tamper-resistent laser-micrometer electronics enclosure and catheter-holding measurement fixture
- Designed and iterated UV-curing fixtures to contain UV exposure, eliminating protective eyewear in work areas
- Identified redundant and non-applicable lot-release tests, eliminating weeks of duplicate test effort

BMW MANUFACTURING
January 2025 - May 2025
Product Process Planning Co-Op | Spartanburg, SC

- Developed a cross functional material-flow plan to support the launch of a new vehicle-processing mezzanine
    - Coordinated twice daily deliveries of 2,000+ parts across logistics, production, and paint-shop teams
- Conducted time-and-motion studies with production team to develop 3 future-state process sheets, supporting throughput growth from 35 to 65 units/hour
- Configured and deployed an onboarding chatbot on the company's internal AI platform
    - Streamlined onboarding for a 20-person team across two offices
    - Reduced repeat questions during intern onboarding cycles (every 5 months)
- Analyzed campus shuttle operations using route data and proposed a new route to cut fuel use by 18%, presenting recommendations to logistics team

PROJECTS

NORTHROP GRUMMAN-SPONSORED SENIOR PROJECT **September 2025 - June 2026**
Manufacturing Engineering Capstone | Cal Poly

- Co-developed augmented reality assembly and inspection templates with a 3 person team using HoloLens 2 and WorkLink for stud placement, stud verification, wire-harness routing, and bolt-torque sequencing
- Validated AR templates through 15 third-party user trials, collecting positional-accuracy, task-completion, and usability data; achieved an 86.6% template-completion rate)

CAL POLY RACING - FORMULA SAE **June 2023 - June 2024**
Battery Systems Lead | Cal Poly

- Designed a specialized enclosure for 480 battery cells, integrating thermal management and data acquisition while strictly adhering to regulatory separation requirements for high and low voltage electronics
- Manufactured and assembled the accumulator, including carbon fiber structure, polycarbonate cell carriers, and integration of electrical components

    - Assisted with the carbon fiber layup of panels
- Led the Critical Design Review for the tractive system, defending component layout, manufacturability, and serviceability to management and alumni

SELECTED MANUFACTURING ENGINEERING COURSE PROJECTS | *Cal Poly*

- **Injection-Molded Screwdriver Handle Tooling:** Designed, CAM-programmed, and machined a two-piece screwdriver-handle mold on a Haas VF-2; incorporated draft, runner/gate, Morgan Press mounting, and nozzle-clearance features, then molded finished handles
- **Bluetooth Speaker Fabrication:** Fabricated and assembled a Bluetooth speaker by soldering electronic components and producing a laser-cut acrylic enclosure.
- **Business Card Holder CNC Machining:** Manually programmed, simulated, and machined business-card holders on a Haas VF-2; set tool and work offsets and completed first-article CMM inspection on Zeiss Durmax CMM
- **Utility Blade CNC Fixture:** Designed, machined, and tested a quick-change CNC fixture for a Haas EC-400 flexible manufacturing system, meeting a ≥3.0 safety factor and <1-minute part-changeover target
- **Medallion Sand Casting:** Designed a medallion pattern and generated CAM toolpaths to machine it in wax for producing a sodium-silicate sand mold

SKILLS AND CERTIFICATIONS

Software: SolidWorks, JMP, Minitab, Siemens NX, Jira, Confluence, Microsoft Office Suite (Word, Excel, PowerPoint)

Prototyping/Fabrication: 3D Printing (SLA, FDM), Mill, Lathe, Laser Cutter, Basic Welding, CNC/CMM Operations, Sand Casting

Certifications: Lean Six Sigma Green Belt, Certified SOLIDWORKS Associate (CSWA)